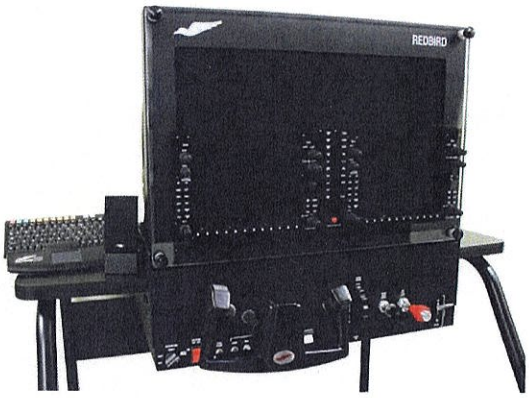
REDBIRD FLIGHT SIMULATIONS

QUALIFICATION AND APPROVAL GUIDE (QAG)

TD / TD2 BASIC AVIATION TRAINING DEVICES
QUALIFICATION AND APPROVAL GUIDE
VERSION 3.0C





Basic Aviation Training Device(s)

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September 25, 2018

FAA APPROVED QAG
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LOG OF REVISIONS

Description of Changes	Version	Edited By
Combined QAGs TD BATD QAG v2.2 and TD2 BATD QAG v2.2	3.0	WJ
Added Navigator Instructor Station in Section 2	3.0	WJ
Added E-Trim, AP, PTT switches to Yoke controller in Section 2 and 3	3.0	WJ
Added TD Platform to Component List in Section 2 and 3	3.0	WJ
Added RealNav to Component List in Section 3	3.0	WJ
Added Windows 10 to Component List in Section 3	3.0	WJ
Added IS PC or Wireless Mobile Device to Component List in Section 3	3.0	WJ
Updated Section 4 with verbatim language from AC 61-136A	3.0	WJ
Added Performance Table in Section 5	3.0	WJ
Added Visual reference images to Section 6	3.0	WJ
Added Section 7: ATD Functions and Maneuvers Checklist	3.0	WJ
Updated GLS capability to “No” in Section 7	3.0A	WJ
Corrected typo on Page 7	3.0B	WJ
Removed reference to AATD in Section 4	3.0B	WJ
Updated Aircraft Configurations in Section 5	3.0B	WJ
Updated MGW in Section 5	3.0B	WJ
Edited language in paragraph a.(2) in Section 4	3.0C	WJ
Updated Performance Table in Section 5	3.0C	WJ
Updated aircraft descriptions with retractable gear enabled in Section 5	3.0C	WJ

SECTION 1: Compliance Statement

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities for the Redbird TD and TD2 models. This includes any optional aircraft configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136A / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: Aviation Training Device (ATD) Description and Pictures

The Redbird TD and TD2 models are based on the general layout of a fixed high-wing aircraft. These models closely represent the overall functionality, performance, avionics, and instrumentation of an industry standard flight training aircraft. The platform consists of standard flight controls, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations.

The Redbird TD and TD2 simulators provide realistic avionics and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

The Redbird TD and TD2 simulators are versatile, expandable and affordable devices that have been designed to be representative of popular high-wing single engine piston aircraft. They offer the user the ability to quickly change the avionics configuration to represent a traditional analog instruments or a glass panel PFD/MFD display.

Redbird TD

- Primary visuals provided by 27” LCD monitor. See Figure 1, 2, 25
- Single pilot control with realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 6, 7, 9, 11
- An interchangeable instrument panel to allow a quick-change options between analog and glass panel configurations. See Figures 12, 13
- Optional monitor configuration to improve visual display capabilities. See Figure 14, 26



Figure 1: Redbird TD

Redbird TD2

- Primary visuals provided by 27” LCD monitor. See Figure 1, 2, 25
- Single pilot control with realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to represent the family of aircraft. See Figures 6, 7, 9, 10
- An interchangeable instrument panel to allow a quick-change options between analog and glass panel configurations. See Figures 12, 13
- User configurable options to enable – Retractable Gears, Variable Pitch Propeller, High Performance Engine. Figure 10
- Optional monitor configuration to improve visual display capabilities. See Figure 14, 26



Figure 2: Redbird TD2

Configuration Components

Instructor's Station

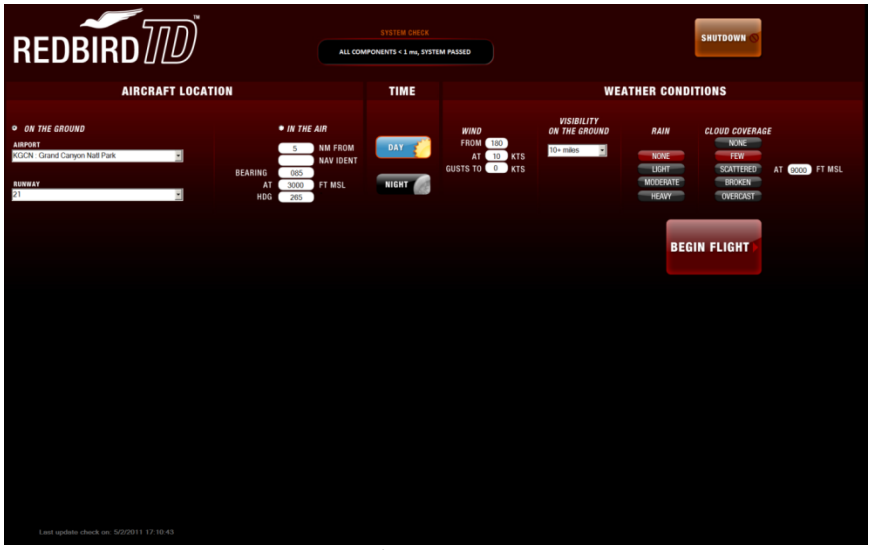


Figure 3: TD/TD2 Session Start Screen

The Redbird TD and TD2 comes standard with an industry standard keyboard to initialize training sessions and provide basic Instructor functions.



Figure 4: Instructor Hotkeys

The optional Navigator full-featured Instructor Station is also available for the Redbird TD and TD2 and is operated from any PC and/or web browser enabled device.

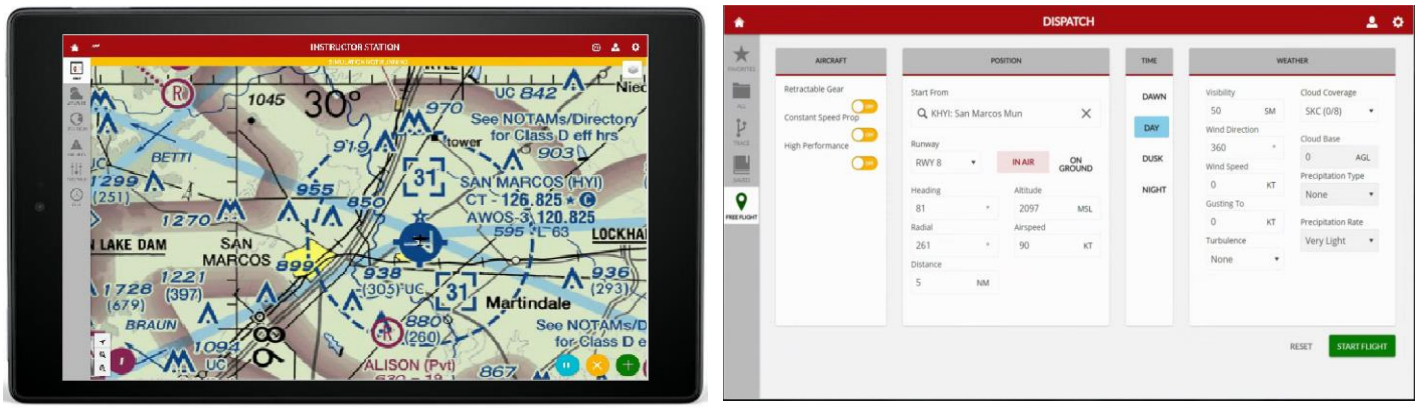


Figure 5: Navigator Instructor Station Interface (web-based)

Controls

The flight controls for Redbird TD and TD2 are realistic in appearance and function and represents flight controls found in many popular aircrafts used for flight training.



Figure 6: Control Yoke (Legacy)



Figure 7: Control Yoke w/ E-Trim, AP, PTT Switches



Figure 8: E-Trim, AP, PTT



Figure 9: Switch Panel



Figure 10: TD2 Throttle Quadrant (T-P-M), Trim, Gear Handle, Flap Handle

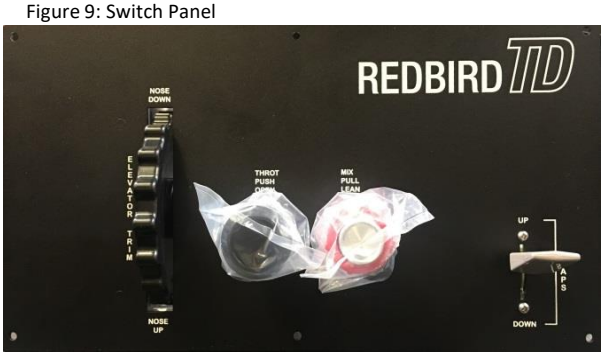


Figure 11: TD Throttle Quadrant (T-P), Trim, Flap Handle

Aircraft Instrument Configurations

The modular design of the Redbird VTO allows for quick configuration changes between different aircrafts. Reconfiguring the VTO for different aircraft configuration is completed by switching the acrylic instrument panel.



Figure 12: Analog Panel



Figure 13: Glass Panel



Figure 14: TD2 - Glass Panel General Overview (Shown with optional Redbird Horizon expanded visuals)



Figure 15: TD2 – Analog Panel General Overview (Shown with optional Redbird Horizon expanded visuals)



Figure 16: TD – Glass Panel General Overview (Shown with optional Redbird Horizon expanded visuals)



Figure 17: TD – Analog Panel General Overview (Shown with optional Redbird Horizon expanded visuals)

Misc. Accessories



Figure 18: TD Platform with optional Horizon Visual Expansion

SECTION 3: Training Device Component List

Qty	Type	Manufacturer	Name	Description/Function	Applicable Configurations
1	Software	Redbird Flight Simulations	RBSim	Simulation control and component integration.	All
1	Software	Redbird Flight Simulations	Navigator	Simulation control and component integration. Environmental, location and failure controls with map, track and glideslope display.	Varies by S/N
1	Software	Microsoft	Windows XP	Operating System	Varies by S/N
1	Software	Microsoft	Windows 7	Operating System	Varies by S/N
1	Software	Microsoft	Windows 10	Operating System	Varies by S/N
1	Software	Microsoft	ESP or FSX	Simulation Engine (includes Map, Track, and Glideslope tracking)	Varies by S/N
1	Software	Lockheed Martin	Prepar3d	Simulation Engine (includes Map, Track, and Glideslope tracking)	Varies by S/N
1	Software	Redbird Flight Simulations	Redbird Analog Gauges	Analog Instruments, Hobbs, Flap and Trim Indicators, Gear Position	See Configuration
1	Software	Mindstar Aviation	Redbird 430/ 530	Virtual GPS and Radios	See Configuration
1	Software	Mindstar Aviation	Redbird Autopilot	Virtual Autopilot (Redbird 140, 55, or 700)	See Configuration
1	Software	Mindstar Aviation	Redbird 1000	Virtual flight instruments, GPS, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Flight 1	Compass and Analog Gauges	Compass, Miscellaneous Gauges	All
1	Software	Redbird Flight Simulations	Parrot and Cygnus	Software providing ATC and location services	All
1	Software	RealNav Data	RealNav Data	Navigation db for simulator	All
1	Hardware	27" Industry Standard	LCD	Flat Panel displays for exterior views and virtual instruments	All
1	Hardware	Redbird Flight Simulations	Pro and RD1 Rudder Pedals	USB Rudder Pedals	All
1	Hardware	CH Products Corporation	Pro Pedals	USB Rudder Pedals	All
1	Hardware	Mad Catz Inc./Saitek	Pro Flight Rudder Pedals	USB Rudder Pedals	All
1	Hardware	Mad Catz Inc./Saitek	Pro Flight Cessna Rudder Pedals	USB Rudder Pedals	All
1	Hardware	Industry Standard	Keyboard and trackpad	USB keyboard and trackpad	All
1	Hardware	Industry Standard	Instructors Station Computer or Wireless Mobile Device	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Redbird TD/TD2 CPU	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	Varies by S/N
1	Hardware	Redbird Flight Simulations	Redbird TD/TD2 Yoke (Legacy)	Yoke Control	Varies by S/N
1	Hardware	Redbird Flight Simulations	Redbird TD/TD2 Yoke	Yoke Control w/ AP, PTT, E-Trim switches	Varies by S/N
1	Hardware	Redbird Flight Simulations	Redbird Throttle Quadrant – TD	Throttle Quadrant	TD Only
1	Hardware	Redbird Flight Simulations	Redbird Throttle Quadrant – TD2	Throttle Quadrant	TD2 Only
1	Hardware	Redbird Flight Simulations	Redbird Instrument Controls Overlay	Virtual Instrument Controls Overlay	See Configuration
1	Hardware	Redbird Flight Simulations	Redbird TD/TD2 Switch Panel Block	Lower Panel Switch Block	All
1	Hardware	Redbird Flight Simulations	Redbird Horizon	Expanded Exterior Visuals	All
1	Hardware	Redbird Flight Simulations	TD Platform	Platform for TD/TD2 (Table, FMX Rudder Pedals, Pilot Seat, Floor)	All

Table 1: Training Device Component List (Above)

Statements of Compatibility of Software and Hardware

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware as noted in AC 61-136A, Appendix 2.

Microsoft Windows (Operating System)
"This is to certify that Microsoft Corporation, the owner and developer of the Windows operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Microsoft ESP or FSX (Simulation Engine)
This is to certify that Microsoft Corporation, the owner and developer of the ESP and FSX Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Lockheed Martin Prepar3D (Simulation Engine)
This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Mindstar Aviation Redbird G430 (Virtual Instrumentation)
This is to certify that Mindstar Aviation, the owner and developer of the Redbird Virtual Instrument Suite, including the Redbird 1000, Redbird 430/530, Redbird 140 Autopilot, Redbird 55 Autopilot, Redbird Entegra and Redbird Perspective along with Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Redbird Virtual Instrument Suite software package is fully compatible with the Redbird Flight Simulations TD and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

Flight 1 Compass
This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations TD and all configurations of that model. Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

SECTION 4: Aviation Training Device (ATD) Design Criteria List

The following section provides the detailed "word for word" listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136A, for BATD requirements Appendix 2) and operational performance value/scale (as applicable) for each of the functions described for the Redbird TD and TD2.

Basic ATD Requirements List [Appendix 2 items]

All configurations for this model, as noted, meet AC 61-136A, Appendix 2 requirements

a. The Redbird TD and TD2 models meets the following Control Input requirements:

(1) Physical flight and aircraft system controls should be recognizable as to their function and how they are to be manipulated solely from their appearance. Physical flight and aircraft system controls eliminate the use of interfaces such as a keyboard, mouse, or gaming joystick to control the represented aircraft model in simulated flight.

(2) virtual control is any input device to control aspects of the simulation (such as setting aircraft configuration, location, and weather) and to program, pause, or freeze the device.

(3) Except for setup and/or fault mode entry, neither the keyboard nor the mouse may be used to set or position any feature of the BATD in the represented aircraft for the maneuvers or flight training to be accomplished. See the control requirements listed below for necessary equipment as applicable to the aircraft model represented. The pilot must operate the additional equipment needed in order to accomplish a training procedure as listed in this appendix in the same manner in which it would be operated in the actual aircraft. This would include the landing gear, wing flaps, cowl flaps, and carburetor heat, mixture, propeller, and throttle controls.

(4) The physical arrangement, appearance, and operation of controls, instruments, and switches required by this appendix should closely model the aircraft in the family of aircraft represented. Manufacturers are expected to recreate the appearance, arrangement, operation, and function of realistically placed physical switches and other required controls representative of an aircraft instrument panel that includes at least the following:

- Master/battery;
- Magnetos for each engine (as applicable);
- Alternators or generators for each engine;
- Auxiliary power unit (APU) (if applicable);
- Fuel boost pumps/prime boost pumps for each engine;
- Avionics master;
- Pitot heat; and
- Rotating beacon/strobe, navigation, taxi, and landing lights.

(5) When an FAA-approved BATD is in use, only the software package evaluated and approved by the FAA may be loaded for use on that system to avoid negative impact on available system resources.

b. The Redbird TD and TD2 models meets the following Control Requirements:

(1) Airplane.
   (a) A self-centering displacement yoke or control stick that allows continuous adjustment of pitch and bank.
   (b) Self-centering rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.
   (c) Throttle or power control(s) that allows continuous movement from idle to full-power settings and corresponding changes in pitch and yaw, as applicable.
   (d) Mixture/condition, propeller, and throttle/power control(s) as applicable to the aircraft or family of aircraft represented.
   (e) Controls for the following items, as applicable to the category and class of aircraft represented:
      - Wing flaps,
      - Pitch trim,
      - Communication and navigation radios,
      - Clock or timer,
      - Gear handle (if applicable),
      - Transponder,
      - Altimeter,
      - Carburetor heat (if applicable), and
      - Cowl flaps (if applicable).

c. The Redbird TD and TD2 models meets the following Control Input requirements:

(1) Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). The manufacturer listed in the qualification and approval guide submitted for FAA approval must verify this performance and capability.
(2) The control inputs must be tested by the computer and software at each startup and displayed as a confirmation message or a warning message that the transport delay time or any design parameter is out of original tolerances. It should not be possible to continue the training session unless the problem is resolved and all components are functioning properly. This test should consider the items listed under “Display Requirements” in subparagraph 3d below.

d. The Redbird TD and TD2 models meets the following Display Requirements:

(1) Instruments and indicators replicated and properly located as appropriate to the aircraft represented:

   (a) Flight instruments in a standard configuration representing the traditional “round” flight instruments. An electronic primary flight display (PFD) with reversionary and backup flight instruments will also be acceptable.
   (b) A sensitive altimeter with incremental markings each 20 feet or less, operable throughout the normal operating range of the aircraft or family of aircraft represented.
   (c) A magnetic direction indicator.

(d) A heading indicator with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the aircraft or family of aircraft represented.
(e) An airspeed indicator with incremental markings as shown on the aircraft or family of aircraft represented; airspeed markings of less than 40 knots need not be displayed.
(f) A vertical speed indicator with incremental markings each 100 fpm for both climb and descent, for the first 1,000 feet per minute (fpm) of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ±2,000 fpm total display, or as applicable to the aircraft or family of aircraft being represented.
(g) A gyroscopic rate-of-turn indicator or equivalent with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.
(h) A slip and skid indicator with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the center position. A split image triangle indication may be used if applicable to the aircraft or family of aircraft being represented.
(i) An attitude indicator with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to the aircraft or family of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.
(j) Engine instruments as applicable to the aircraft or family of aircraft being represented, providing markings for normal ranges and minimum and maximum limits.
(k) A suction gauge or instrument pressure gauge with a display applicable to the aircraft represented.
(l) A flap setting indicator that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.
(m) A pitch trim indicator with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.
(n) Communication radio(s) with display(s) of the radio frequency in use.
(o) Navigation radio(s) capable of replicating both precision and nonprecision instruments, including approach procedures (each with an aural identification feature), and a marker beacon receiver. For example, an instrument landing system (ILS), non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or Very high frequency Omnidirectional Range (VOR). Graduated markings as indicated below must be present on each course deviation indicator (CDI) as applicable. The marking should include:
    - One-half dot or less for course/glideslope (GS) deviation (i.e., VOR, LOC, or ILS), and
    - Five degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable.
(p) A clock with incremental markings for each minute and second, or a timer with a display of minutes and seconds.
(q) A transponder that displays the current transponder setting.
(r) A fuel quantity indicator(s) that displays the fuel remaining, either in analog or digital format, appropriate for the aircraft or family of aircraft represented.

(2) All instrument displays listed above must be visible during all flight operations. Allowances can be made for multifunction electronic displays that may not display all instruments simultaneously. The update rate of all displays must provide an image of the instrument that:

(s) Does not appear to be out of focus or illegible
(t) Does not appear to "jump" or "step" to a distracting degree during operation.
(u) Does not appear with distracting jagged lines or edges.
(v) Does not appear to lag relative to the action and use of the flight controls.

(3) Control inputs should be reflected by the flight instruments in real time and without a perceived delay in action. Display updates must display all changes (within the total range of the replicated instrument) that are equal to or greater than the values stated below:
(a) Airspeed indicator: change of 5 knots.
(b) Attitude indicator: change of 2 degrees in pitch and bank.
(c) Altimeter: change of 10 feet.
(d) Turn and bank: change of ¼ standard rate turn.
(e) Heading indicator: change of 2 degrees.
(f) Vertical speed indicator (VSI): change of 100 fpm.
(g) Tachometer: change of 25 rpm or 2 percent of turbine speed.
(h) VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.
(i) ADF: change of 2 degrees.
(j) GPS: change as appropriate for the model of GPS based navigator represented.
(k) Clock or timer: change of 1 second.

(4) Displays must reflect dynamic behavior of an actual aircraft display (e.g., a VSI reading of 500 fpm must reflect a corresponding movement in altimeter and an increase in power must reflect an increase in the rpm indication or power indicator.)

e. The Redbird TD and TD2 models meets the following Flight Dynamics Requirements:

(1) Flight dynamics of the ATD should be comparable to the way the represented training aircraft performs and handles. However, there is no requirement for an ATD to have control loading to exactly replicate any particular aircraft. An air data-handling package is not required for determination of forces to simulate during the manufacturing process.
(2) Aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate, and hovering/sideward/forward/rearward flight) should be comparable to the aircraft or family of aircraft being represented. A performance table will need to be included in the QAG for each aircraft configuration for sea level and 5,000 feet.
(3) Aircraft vertical lift component must change as a function of bank comparable to the way the aircraft or family of aircraft being represented performs and handles.
(4) Changes in flap setting, slat setting, gear position, collective control, or cyclic control must be accompanied by changes in flight dynamics comparable to the way the aircraft or family of aircraft represented performs and handles.
(5) The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft or family of aircraft represented performs and handles.

f. The Redbird TD and TD2 models meets the following Instructional Management Requirements:

(1) The instructor must be able to pause the system at any point for the purpose of administering instruction regarding the task.
(2) If a training session begins with the "aircraft in the air" and ready for the performance of a particular procedural task, the instructor must be able to manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

(3) The system must be capable of recording both a horizontal and vertical track of aircraft movement during the entire training session for later playback and review.

(4) The instructor must be able to disable any of the instruments prior to or during a training session and be able to simulate failure of any of the instruments without stopping or freezing the simulation to affect the failure.

(5) The ATD must have at least a navigational area database that is local to the training facility to allow reinforcement of procedures learned during actual flight in that area. All navigational data must be based on procedures as published per 14 CFR part 97.

SECTION 5: Aircraft Configurations

List of Previously Approved Configurations (QAG v2.2):

TD – C172 ANALOG



Figure 19: TD C172 - Analog

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 11)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**

TD – C172 GLASS



Figure 20: TD C172 – Glass

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 11)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

TD2 – C172 ANALOG



Figure 21: TD2 C172 – Analog

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**

TD2 – C172RG ANALOG



Figure 22: TD2 C172 – Analog & Retractable Gears

MTOW	FWD/AFT CG Limit
2650 lbs	39.5 in / 46.5 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Retractable gears**

TD2 – C172 CSP ANALOG



Figure 21: TD2 C172 – Analog & CS Prop

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Constant Speed Propeller Control**

TD2 – C172 CSP RG ANALOG



Figure 21: TD2 C172 – Analog, CS Prop & Retractable Gears

MTOW	FWD/AFT CG Limit
2650 lbs	39.5 in / 46.5 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Constant Speed Propeller Control**
- **Retractable gears**

TD2 – C172 GLASS



Figure 23: TD2 C172 – Glass

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

TD2 – C172 RG GLASS



Figure 24: TD2 C172 – Glass & Retractable Gears

MTOW	FWD/AFT CG Limit
2650 lbs	39.5 in / 46.5 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Retractable gears**

TD2 – C172 CSP GLASS



Figure 23: TD2 C172 – Glass & CS Prop

MTOW	FWD/AFT CG Limit
2550 lbs	41 in / 47.3 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Constant Speed Propeller Control**

TD2 – C172 CSP RG GLASS



Figure 24: TD2 C172 – Glass, CS Prop & Retractable Gears

MTOW	FWD/AFT CG Limit
2650 lbs	39.5 in / 46.5 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Constant Speed Propeller Control**
- **Retractable gears**

TD2 – C182 ANALOG



Figure 25: TD2 C182 – Analog

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**

TD2 – C182 RG ANALOG



Figure 25: TD2 C182 – Analog & Retractable Gears

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Retractable gears**

TD2 – C182 CSP ANALOG



Figure 25: TD2 C182 – Analog & CS Prop

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Constant Speed Propeller Control**

TD2 – C182 CSP RG ANALOG



Figure 26: TD2 C182 – Analog, CS Prop & Retractable Gears

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Analog gauges & 530 GPS with KAP140 Autopilot**
- **Constant Speed Propeller Control**
- **Retractable gears**

TD2 – C182 GLASS



Figure 27: TD2 C182 – Glass

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**

TD2 – C182 RG GLASS



Figure 27: C182 – Glass & Retractable Gears

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Retractable gears**

TD2 – C182 CSP GLASS



Figure 27: C182 – Glass & CS Prop

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Constant Speed Propeller Control**

TD2 – C182 CSP RG GLASS



Figure 28: C182 – Glass, CS Prop & Retractable Gears

MTOW	FWD/AFT CG Limit
3100 lbs	40.9 in / 46 in

- **Yoke – Center (See Figures 6,7,8)**
- **Throttle – Single Engine Vernier (T-P-M) (See Figure 10)**
- **Glass Cockpit G1000 PFD/MFD & GFC700 Autopilot**
- **Constant Speed Propeller Control**
- **Retractable gears**

Additional configurations included in this version (QAG v3.0C)

None

Performance Table

Aircraft Model	V_{SO}	V_{S1}	V_X	V_Y	V_A	V_{NE}	V_{MCA}	KTAS @ Cruise / 75% power setting	Rate of Climb (fpm) @ (V_Y) / Full Power	Single Engine Rate of Climb (V_{YSE})
TD – C172	40 KIAS	44 KIAS	62 KIAS	74 KIAS	105 KIAS	163 KIAS	N/A	114 KTAS**	1125 fpm**	N/A
							5000' -->	120 KTAS	890 fpm	N/A
TD2 – C172	40 KIAS	44 KIAS	62 KIAS	74 KIAS	105 KIAS	163 KIAS	N/A	114 KTAS**	1125 fpm**	N/A
							5000' -->	120 KTAS	890 fpm	N/A
TD2 – C172CSP	40 KIAS	44 KIAS	62 KIAS	74 KIAS	105 KIAS	163 KIAS	N/A	120 KTAS**	1190 fpm**	N/A
							5000' -->	125 KTAS	900 fpm	N/A
TD2 – C172RG	42 KIAS*	47 KIAS*	62 KIAS	84 KIAS	106 KIAS	164 KIAS	N/A	120 KTAS**	1200 fpm**	N/A
							5000' -->	126 KTAS	920 fpm	N/A
TD2 – C172CSPRG	42 KIAS*	47 KIAS*	62 KIAS	84 KIAS	106 KIAS	164 KIAS	N/A	124 KTAS**	1230 fpm**	N/A
							5000' -->	132 KTAS	945 fpm	N/A
TD2 – C182	50 KIAS	56 KIAS	63 KIAS	80 KIAS	110 KIAS	175 KIAS	N/A	117 KTAS**	810 fpm**	N/A
							5000' -->	123 KTAS	620 fpm	N/A
TD2 – C182CSP	50 KIAS	56 KIAS	63 KIAS	80 KIAS	110 KIAS	175 KIAS	N/A	120 KTAS**	825 fpm**	N/A
							5000' -->	127 KTAS	625 fpm	N/A
TD2 – C182RG	50 KIAS*	56 KIAS*	65 KIAS	88 KIAS	112 KIAS	181 KIAS	N/A	119 KTAS**	900 fpm**	N/A
							5000' -->	129 KTAS	740 fpm	N/A
TD2 – C182CSPRG	50 KIAS*	56 KIAS*	65 KIAS	88 KIAS	112 KIAS	181 KIAS	N/A	128 KTAS**	900 fpm**	N/A
							5000' -->	136 KTAS	750 fpm	N/A

*MOST FORWARD CENTER OF GRAVITY

**2000 FEET

Table 2: Aircraft Performance

SECTION 6: Visual System with VFR, IFR, DAY, and NIGHT Capabilities

Redbird TD/TD2 Visual System
The TD/TD2 visual system is capable of providing the following flight conditions:

Visual Flight Rules
- Day – See Figure 31
- Night – See Figure 32

Instrument Flight Rules
- Day – See Figure 33
- Night – See Figure 34

The flight condition is set by the user or instructor from the Session Start Screen (Figure 3, 29) and can be adjusted during the training session by the user/instructor from the Keyboard shortcut (Figure 4) as well as the Navigator Instructor Station (Figure 5, 30).



Figure 29: Session Start Screen Weather Settings

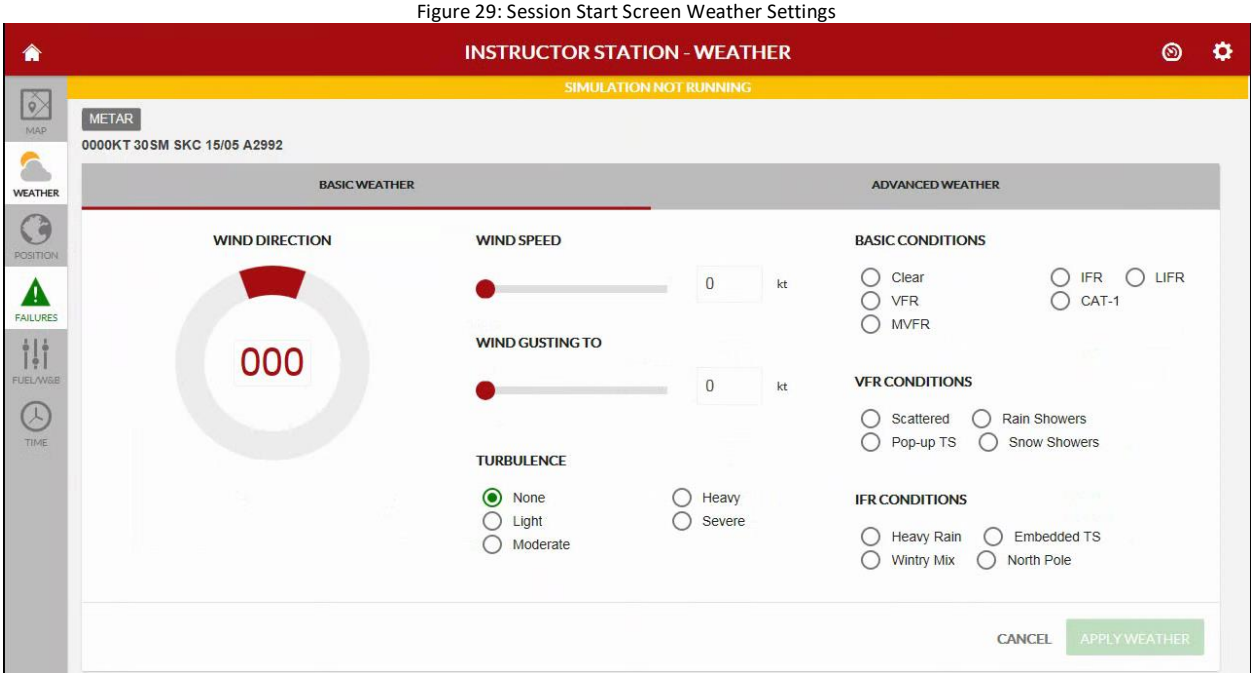


Figure 30: Navigator Instructor Station Weather Settings

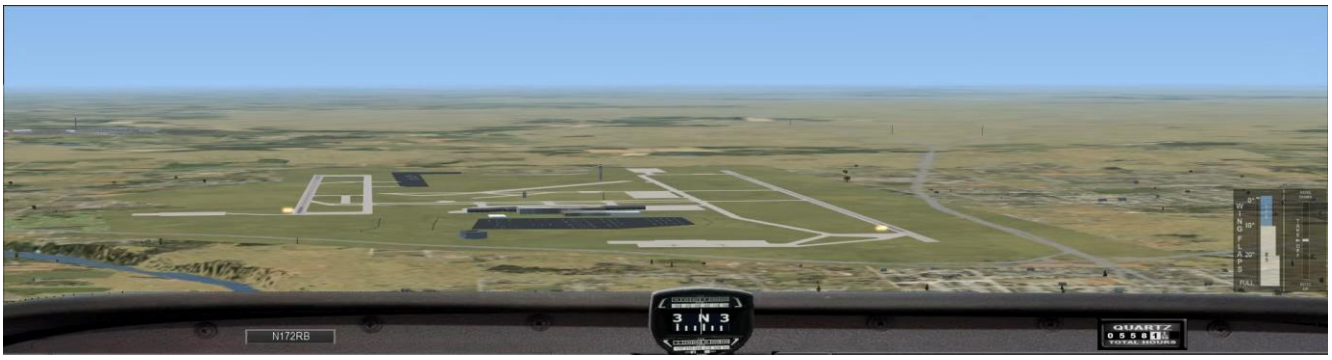


Figure 31: VFR Day



Figure 32: VFR Night



Figure 33: IFR Day



Figure 34: IFR Night

The Redbird Horizon (optional expanded visuals) extends the TD/TD2 visual system with two additional monitors to provide extended visual referencing.



Figure 35: Standard TD/TD Visual System



Figure 36: TD/TD2 with optional Redbird Horizon Visual System

SECTION 7: ATD Functions and Maneuvers Checklist

ATD FUNCTIONS AND MANEUVERS CHECKLIST

Functions and Maneuvers	Satisfactory? Yes/No
a. Preparation for Flight	
(1) Preflight	Yes
b. Pre-Takeoff	
(1) Engine start	Yes
(2) Brake operation	Yes
c. Takeoff	
(1) **AIRPLANE Takeoff**	
(i) Power plant checks	Yes
(ii) Acceleration characteristics	Yes
(iii) Nose wheel and rudder steering	Yes
(iv) Effect of crosswind	Yes
(v) Instrument	Yes
(vi) Landing gear, wing flap operation	Yes
(2) **HELICOPTER Takeoff**	
(i) Power plant checks	N/A
(ii) From Hover	N/A
(iii) From Ground	N/A
(iv) Vertical	N/A
(v) Running	N/A
d. In-flight Operation	
(1) **AIRPLANE In-flight Operation**	
(i) Climb	Yes
(a) Normal	Yes
(b) One engine inoperative procedures	N/A
(ii) Cruise	
(l) Performance characteristics (speed vs. power)	Yes
(b) Normal and steep turns	Yes
(c) Performance turns	Yes
(d) Approach to stalls, i.e., stall warning (cruise, takeoff/approach, and landing configuration)	Yes
(e) High angle of attack maneuvers (cruise, takeoff/approach, and landing)	Yes
(f) In-flight engine shutdown	Yes
(g) In-flight engine restart	Yes
(iii) Descent	Yes
(a) Normal	Yes
(b) Maximum rate	Yes
(2) **HELICOPTER In-flight Operation**	
(i) Hovering	N/A
(a) Forward	N/A
(b) Rearward	N/A
(c) Sideward	N/A
(d) Turns	N/A
(ii) Climb	N/A
(iii) Cruise	N/A
(a) Performance characteristics	N/A
(b) Turns	N/A
i. Recovery	N/A

ii. Skidding	N/A
iii. Slipping	N/A
(c) In-flight engine shutdown	N/A
(d) In-flight engine restart	N/A
(iv) Descent	N/A
e. Approaches	
(1) Non-precision	Yes
(i) GPS-WAAS	Varies*
(ii) GPS-LAAS	Varies*
(iii) All engines operating	Yes
(iv) One or more engines inoperative	N/A
(v) Approach procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS, & LPV, and additional types when developed and certified)	Yes
vi) Missed approach (all engines operational, one or more engines inoperative)	Yes
(2) Precision	Yes
(i) PAR – normal	Yes
(ii) ILS	Yes
(iii) GLS	N/A
(iv) Effects of crosswind	Yes
(v) With engine(s) inoperative	Yes
(vi) Missed approach	Yes
(a) Normal	Yes
(b) With engine(s) inoperative	Yes
(c) From steep glide slope	Yes
f. Surface Operations (AIRPLANE-Post Landing)	
(1) Landing roll	Yes
(2) Braking operation	Yes
(3) Reverse thrust operation, if applicable	N/A
g. Any Flight Phase	
(1) Aircraft and power plant systems	Yes
(i) Electrical	Yes
(ii) Flaps (**AIRPLANE**)	Yes
(iii) Fuel and oil	Yes
(iv) Landing gear (**AIRPLANE**)	Yes
(2) Flight management and guidance systems	Yes
(i) Automatic pilot	Yes
(ii) Flight director/system displays	Yes
(iii) Navigation systems	Yes
(iv) Stall warning/avoidance (**AIRPLANE**)	Yes
(v) Multi-function displays	Varies*
(vi) Primary flight displays	Varies*
(3) Airborne procedures	Yes
(i) Holding	Yes
(4) Engine shutdown and parking	Yes
(i) Systems operation	Yes
(ii) Parking brake operation (**AIRPLANE**)	Yes

*Based on configuration

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.